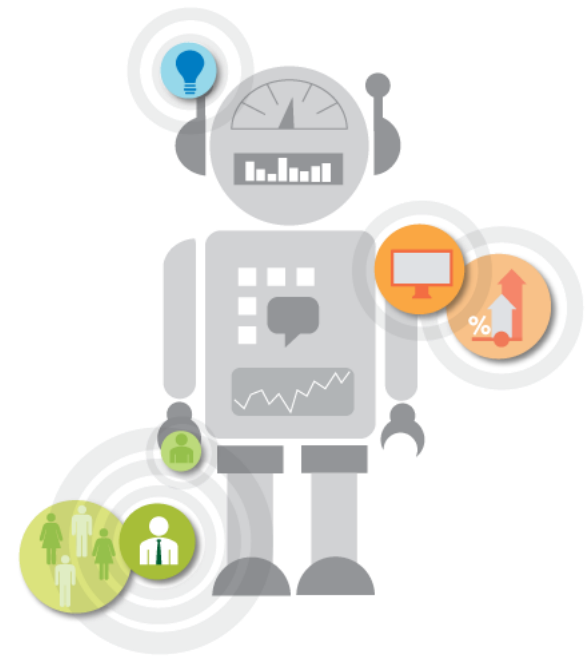


AI and Society

CSET-485

Dr. Susmita Das



UNIT - 2



What is Explainable AI?

Explainable AI (XAI)

Explainable AI (XAI) refers to artificial intelligence systems that are designed so humans can **understand, interpret, and trust** how they make decisions.

Instead of being a “black box” (where inputs go in and outputs come out without clarity), explainable AI provides transparent reasoning behind its predictions or decisions.

Why XAI is Important?

AI systems are increasingly used in high-stakes domains such as:

- Healthcare (diagnosis recommendations)
- Finance (loan approvals)
- Criminal justice (risk assessments)

If an AI denies someone a loan or predicts a disease, stakeholders need to know:

- Why was this decision made and what factors influenced it?
- Can it be challenged or corrected?

Techniques Used in Explainable AI (XAI)

LIME (Local Interpretable Model-agnostic Explanations)

Explains individual predictions by approximating the complex model locally with a simpler interpretable model.

SHAP (SHapley Additive exPlanations)

Based on game theory. It assigns each feature a contribution value to the final prediction.

Grad-CAM

Used in deep learning for computer vision. Highlights image regions influencing a prediction.

Types of Explanations

Global Explanation: Explains how the entire model works overall. *Example:* Age and income are the most important factors in loan approval

Local Explanation: Explains a single prediction. *Example:* This loan was rejected because income was low and credit score was poor

SHAP(SHapley Additive exPlanations)

SHAP is a method in Explainable AI that explains the output of any machine learning model by assigning each feature a **fair contribution value** toward a prediction.

It breaks a prediction into: *Prediction=Baseline + Feature Contributions*

Where:

Baseline = average prediction of the model

Feature contribution (SHAP value) = impact of each feature

How SHAP Works:

1. Treats each feature as a “player” in a cooperative game.
2. Measures how the prediction changes when a feature is added.
3. Averages this contribution across all possible feature combinations.
4. Assigns a fair contribution to each feature.

Maths Behind Shapley Values

Suppose two colleagues enter a Kaggle Contest and win the first prize. This is known as **Coalition value**.

This means, Coalition value of $C_{12} = \$10k$.

When the money is to be split, Player 2 suggests splitting the prize money equally. But Player 1 are not happy with this suggestion, because all the model training was done by Player 1.



$$C_{12} = \$10,000$$

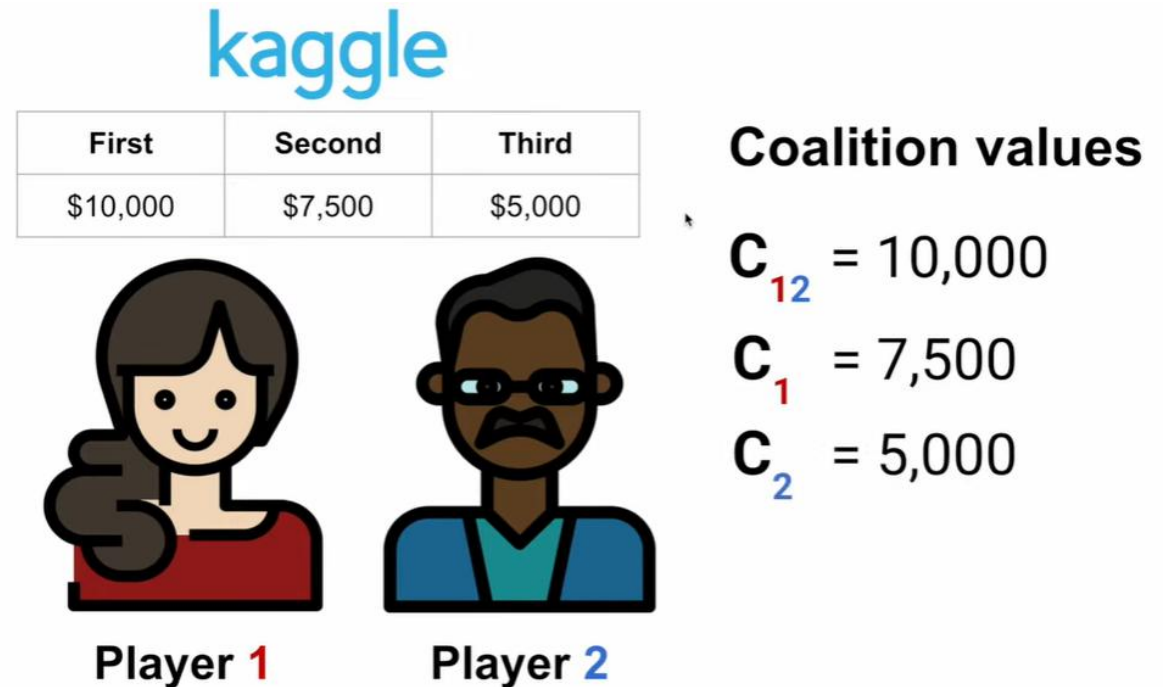
C_{12} is the value of the coalition of player 1 and player 2

Due to this conflict in opinions, both players agree on making a fair estimation of their contributions in the project they built for the Kaggle Contest.

Maths Behind Shapley Values

Both of them re-do the contest alone.
Player 1 comes second while Player 2 comes third.

So, the single coalition value of Player 1,
 $C_1 = \$7.5k$
And the single coalition value of Player 2,
 $C_2 = \$5k$



Although this proves that Player 1 has more contribution in the two-player coalition C_{12} . But this does not solve the problem of how much is the contribution of Player 1 in the joint coalition.

Maths Behind Shapley Values

We calculate the marginal Contribution of each players.

For Player 1:

When Player 1 joins C_2 and make it C_{12} , the coalition value goes from **\$5k to \$10k**.

When Player 1 doesn't join any coalition i.e. C_0 , the coalition value goes from **\$0 to \$7.5k**.

We take the average of both the values which comes out to be **\$6250**

Coalition values

$$C_{12} = 10,000$$

$$C_{12} = 7,500$$

$$C_2 = 5,000$$

$$C_0 = 0$$



$$C_{12} - C_2 = 5,000$$

$$C_{12} - C_0 = 7,500$$

$$(5,000 + 7,500) / 2 = \$6,250$$

Marginal contribution

The increase in a coalition's value due to a player joining that coalition

Maths Behind Shapley Values

For Player 2:

When Player 2 joins C_1 and make it C_{12} , the coalition value goes from **\$7.5k to \$10k**.

When Player 2 doesn't join any coalition i.e. C_0 , the coalition value goes from **\$0 to \$5k**.

We take the weighted average of both the values which comes out to be **\$3750**

Coalition values

$$C_{12} = 10,000$$

$$C_1 = 7,500$$

$$C_2 = 5,000$$

$$C_0 = 0$$

Marginal contribution

The increase in a coalition's value due to a player joining that coalition



$$C_{12} - C_2 = 5,000$$

$$C_1 - C_0 = 7,500$$

$$(5,000 + 7,500) / 2 = \mathbf{\$6,250}$$



$$C_{12} - C_1 = 2,500$$

$$C_2 - C_0 = 5,000$$

$$(2500 + 5000) / 2 = \mathbf{\$3,750}$$

Maths Behind Shapley Values

Adding the expected marginal contribution values of both Player 1 and Player 2:
\$6250+\$3750 = \$10000 gives us the first prize money.

These expected marginal contribution values can be called the **Shapley values** and the money can be split according between the players according to the Shapley values.

Shapley Value Formula (Core Idea) [Formula is explained in details in later slides]

$$\phi_i = \sum_{S \subseteq \{1, \dots, p\} \setminus \{i\}} \frac{|S|!(p-|S|-1)!}{p!} [val(S \cup \{i\}) - val(S)]$$

Maths Behind Shapley Values

Let us a coalition of three players.

The coalition C_{123} wins the first prize of \$10k. Hence, Coalition value of $C_{123} = 10000$



$$C_{123} = 10,000$$
$$C_0 = 0$$

Let us consider, Coalition of Player 1 and Player 2 i.e. C_{12} , wins second prize, then coalition value of $C_{12} = 7500$

Coalition of Player 1 and Player 3 i.e. C_{13} , also wins second prize, then coalition value of $C_{13} = 7500$

$$C_{12} = 7,500$$
$$C_{13} = 7,500$$
$$C_{23} = 5,000$$

Coalition of Player 2 and Player 3 i.e. C_{23} , wins third prize, then coalition value of $C_{23} = 5000$

Individually, Player 1 coalition i.e. $C_1 = 5000$, Player 2 coalition i.e. $C_2 = 5000$ and Player 3 coalition i.e. $C_3 = 0$

$$C_1 = 5,000$$
$$C_2 = 5,000$$
$$C_3 = 0$$

Maths Behind Shapley Values

For Player 1:

Player 1 joining C_{23} , will raise the value from 5000 to 10000.

Player 1 joining C_2 , will raise the value from 5000 to 7500.

Player 1 joining C_3 , will raise the value from 0 to 7500.

If Player 1 doesn't join any coalition, value will remain at **5000**



$$C_{123} = 10,000$$
$$C_0 = 0$$

$$C_{12} = 7,500$$
$$C_{13} = 7,500$$
$$C_{23} = 5,000$$

$$C_1 = 5,000$$
$$C_2 = 5,000$$
$$C_3 = 0$$



$$C_{123} - C_{23} = 5,000$$
$$C_{12} - C_2 = 2,500$$
$$C_{13} - C_3 = 7,500$$

Individually, Player 1 coalition i.e. $C_1 = 5000$, Player 2 coalition i.e. $C_2 = 5000$ and Player 3 coalition i.e. $C_3 = 0$

Maths Behind Shapley Values

For Player 1:

Player 1 joining C_{23} , will raise the value from 5000 to 10000.

Player 1 joining C_2 , will raise the value from 5000 to 7500.

Player 1 joining C_3 , will raise the value from 0 to 7500.

If Player 1 doesn't join any coalition, value will remain at **5000**



$$C_{123} = 10,000$$
$$C_0 = 0$$

$$C_{12} = 7,500$$
$$C_{13} = 7,500$$
$$C_{23} = 5,000$$

$$C_1 = 5,000$$
$$C_2 = 5,000$$
$$C_3 = 0$$



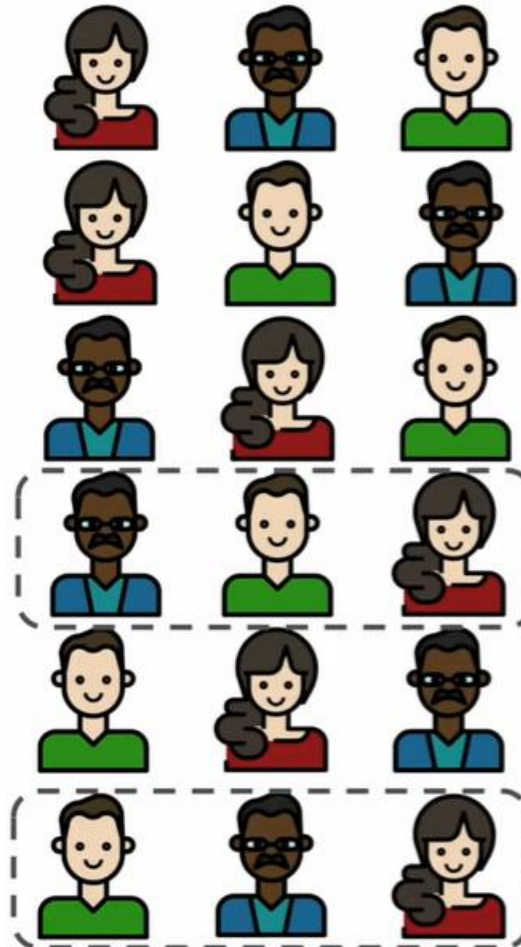
$$C_{123} - C_{23} = 5,000$$
$$C_{12} - C_2 = 2,500$$
$$C_{13} - C_3 = 7,500$$

Individually, Player 1 coalition i.e. $C_1 = 5000$, Player 2 coalition i.e. $C_2 = 5000$ and Player 3 coalition i.e. $C_3 = 0$

Maths Behind Shapley Values

If we consider sequence of joining, there are **6 ways** in which Player 1, Player 2 and Player 3 can make teams.

Among these, there are **2 ways** in which **Player 1 joins the already formed coalition of C_{23}** . Hence, the probability of Player 1 joining C_{23} is $2/6 = 1/3$



$$C_{123} - C_{23} = 5,000$$

$P(C_{123} - C_{23})$ = probability that player 1 makes a marginal contribution to a coalition of player 2 and 3

$$3! = 6$$

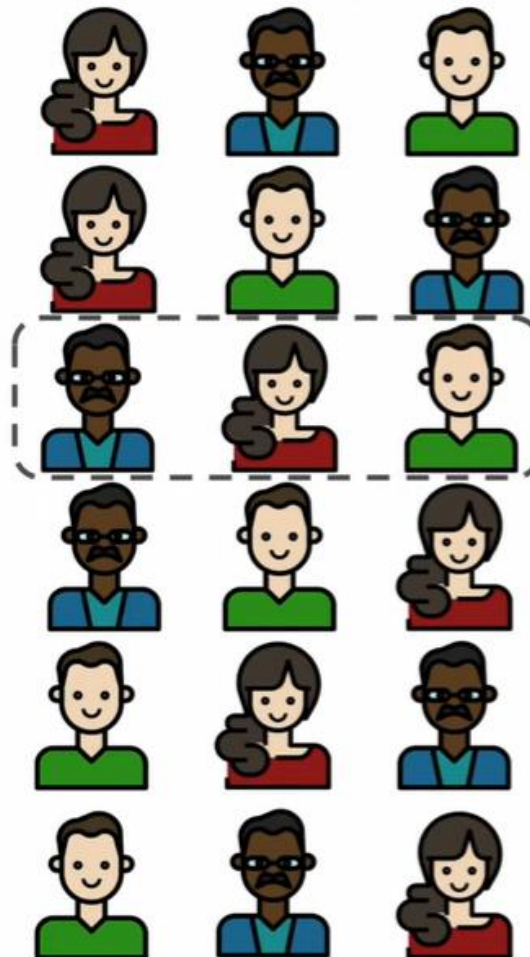
$$2! * 1! = 2$$

$$P(C_{123} - C_{23}) = 2/6 = 1/3$$

Maths Behind Shapley Values

If we consider sequence of joining, there are **6 ways** in which Player 1, Player 2 and Player 3 can make teams.

Among these, there is **only 1 way** in which **Player 1 joins Player 2** and the already formed coalition of C_2 . Hence, the probability of Player 1 joining C_2 is **1/6**



$$C_{123} - C_{23} = 5,000$$

$P(C_{123} - C_{23})$ = probability that player 1 makes a marginal contribution to a coalition of player 2 and 3

$$3! = 6$$

$$2! * 1! = 2$$

$$P(C_{123} - C_{23}) = 2/6 = 1/3$$

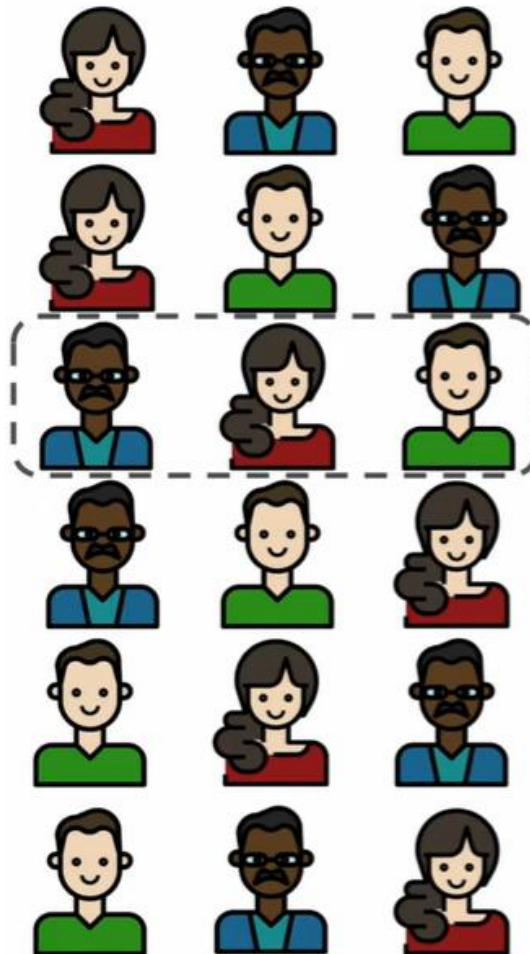
$$P(C_{12} - C_2) = 1/6$$

Maths Behind Shapley Values

If we consider sequence of joining, there are **6 ways** in which Player 1, Player 2 and Player 3 can make teams.

Among these, there is **only 1 way** in which **Player 1 joins Player 3** and the already formed coalition of C_3 . Hence, the probability of Player 1 joining C_3 is **1/6**

There are **2 ways** in which Player 1 does not join any coalition at all.



$$C_{123} - C_{23} = 5,000$$

$P(C_{123} - C_{23})$ = probability that player 1 makes a marginal contribution to a coalition of player 2 and 3

$$3! = 6$$

$$2! * 1! = 2$$

$$P(C_{123} - C_{23}) = 2/6 = 1/3$$

$$P(C_{12} - C_2) = 1/6$$

$$P(C_{13} - C_3) = 1/6$$

$$P(C_1 - C_0) = 2/6 = 1/3$$

Maths Behind Shapley Values

$$\phi_i = \sum_{S \subseteq \{1, \dots, p\} \setminus \{i\}} \underbrace{\frac{|S|!(p-|S|-1)!}{p!}}_{\text{Weight}} \underbrace{[val(S \cup \{i\}) - val(S)]}_{\text{Marginal contribution of player } i \text{ to coalition } S}$$

$p!$ = number of ways to form a coalition of p players

$|S|$ = number of players in coalition S

$|S|!$ = number of ways coalition S can form

$(p-|S|-1)!$ = number of ways players can join after player i joins

Maths Behind Shapley Values

$$P(C_{123} - C_{23}) = 2/6 = 1/3$$

$$P(C_{12} - C_2) = 1/6$$

$$P(C_{13} - C_3) = 1/6$$

$$P(C_1 - C_0) = 2/6 = 1/3$$



$$C_{123} = 10,000$$

$$C_0 = 0$$

$$C_{12} = 7,500$$

$$C_{13} = 7,500$$

$$C_{23} = 5,000$$

$$C_1 = 5,000$$

$$C_2 = 5,000$$

$$C_3 = 0$$



$$C_{123} - C_{23} = 5,000$$

$$C_{12} - C_2 = 2,500$$

$$C_{13} - C_3 = 7,500$$

$$C_1 - C_0 = 5,000$$

$$\begin{aligned} &5,000 * (1/3) + 2,500 * (1/6) \\ &+ 7,500 * (1/6) + 5,000 * (1/3) \\ &= \mathbf{\$5,000} \end{aligned}$$

Shapley Values for Machine Learning

$$\phi_i = \sum_{S \subseteq \{1, \dots, p\} \setminus \{i\}} \frac{|S|!(p-|S|-1)!}{p!} [val(S \cup \{i\}) - val(S)]$$

S is no longer a coalition of players, but **coalition of features in a model**

$$val_x(S) = \int f(x_1, \dots, x_p) dP_{x \notin S}$$

The value function is no longer a value of a game, but **the value predicted by a model**. 'f' is the **function for model prediction** and it has **p features**.

S = coalition of feature values

f = model prediction

p = number of features in the model

This means that, the **value of a coalition of features S**, is the **model's prediction marginalized over all the features that are not in S**. For the features that are in S, we use the actual values

Shapley Values for Machine Learning

$$f(x_1, x_2) = 200x_1 + 1000x_2$$

age $\rightarrow x_1 \in [18, 60]$

degree $\rightarrow x_2 \in \{0, 1\}$



Age: 20

Degree: 1

Predicted
income: \$5,000

Suppose, we want to predict the income of people. We consider two features, age and degree.

Feature x_1 represents **age** which is uniformly distributed between **18 to 60**.

Feature x_2 represents **degree** which shows that there is equal chance that someone will either **have degree** or **not have degree**

Assume, a person having **age 20** and **having a degree** comes along. The model predicts the **income as \$5k**

Shapley Values for Machine Learning

$$f(x_1, x_2) = 200x_1 + 1000x_2$$

$$age \rightarrow x_1 \in [18, 60]$$

$$degree \rightarrow x_2 \in \{0, 1\}$$



Age: 20

Degree: 1

Predicted
income: \$5,000

We want to calculate the marginal contribution of degree {2} to the coalition of age {1}

Start by calculating the value of coalition of both features. S contains both features. So, no need of marginalizing over any features. Instead we use the actual values and the value obtained is same as the model predicted.

$$\begin{aligned} val_x(\{1, 2\}) &= f(20, 1) \\ &= 200(20) + 1000(1) \\ &= 5000 \end{aligned}$$

Shapley Values for Machine Learning

$$f(x_1, x_2) = 200x_1 + 1000x_2$$

$$age \rightarrow x_1 \in [18, 60]$$

$$degree \rightarrow x_2 \in \{0, 1\}$$



Age: 20

Degree: 1

Predicted
income: \$5,000

Next, we have to calculate for the coalition of age{1}. Now, S does not contain degree{2} anymore. We have to marginalize over degree and use actual value for age. Degree is not continuous to marginalize. So, we don't use integration. Instead we use sum the prediction at each value times the probability of that value which for degree{2} is 50%.

$$\begin{aligned} val_x(\{1\}) &= \int f(20, x_2) dP_{x_2} \\ &= \sum_{i=0}^1 f(20, i) P(x_2 = i) \\ &= (200(20) + 1000(0))(0.5) + \\ &\quad (200(20) + 1000(1))(0.5) \\ &= 4500 \end{aligned}$$

$$val_x(\{1, 2\}) - val_x(\{1\}) = 500$$

Importance of Shapley Values

Efficiency: In machine learning, efficiency means the prediction is divided amongst the features. The Shapley values satisfies the equation:

$$f(x) = \sum_{i=1}^p \phi_i + E_X[f(X)]$$

Which means, the sum of all Shapley values and the average predicted values is equal to the prediction.

Symmetry: Two features will have same Shapley value, if they make the same contributions to all coalitions.

Additivity: Shapley values for machine learning are additive